

4.24

$$(a) \cos(\omega_0 n) = \frac{1}{2}(e^{j\omega_0 n} + e^{-j\omega_0 n}),$$

$$1 \xleftrightarrow{DTFT} 2\pi\delta(\omega), \quad (\text{time shifting thm.: } x[n] \xleftrightarrow{DTFT} X(e^{j\omega}), \quad e^{j\omega_0 n} x[n] \xleftrightarrow{DTFT} X(e^{j(\omega-\omega_0)}))$$

$$X(e^{j\omega}) = \frac{1}{2}[2\pi\delta(\omega - \omega_0) + 2\pi\delta(\omega + \omega_0)] = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

(b) Eq. (4.46) gives $H(e^{j\omega}) = \frac{j\omega}{T}$, $|\omega| < \pi$. Then

$$\begin{aligned} Y(e^{j\omega}) &= H(e^{j\omega})X(e^{j\omega}) \\ &= \frac{j\omega}{T}[\pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)] \\ &= \frac{j\omega_0}{T}\pi\delta(\omega - \omega_0) - \frac{j\omega_0}{T}\pi\delta(\omega + \omega_0), \quad |\omega| < \pi. \end{aligned}$$

(c) From Eq. (4.32),

$$\begin{aligned} Y_r(j\Omega) &= H_r(j\Omega)Y(e^{j\Omega T}) = \begin{cases} TY(e^{j\Omega T}), & |\Omega| < \frac{\pi}{T} \\ 0, & \text{otherwise} \end{cases} \\ &= j\omega_0\pi\delta(\Omega T - \omega_0) - j\omega_0\pi\delta(\Omega T + \omega_0) \\ &= j\omega_0\pi\delta[(\Omega - \omega_0/T)T] - j\omega_0\pi\delta[(\Omega + \omega_0/T)T]. \end{aligned}$$

(d) The inverse Fourier transform of $\delta(\Omega T)$ is the constant $1/(2\pi T)$. We then have

$$\begin{aligned} y_r(t) &= \frac{j\omega_0}{2T}e^{j\omega_0 t/T} - \frac{j\omega_0}{2T}e^{-j\omega_0 t/T} \\ &= -\Omega_0 \left(\frac{e^{j\omega_0 t/T} - e^{-j\omega_0 t/T}}{j2} \right) \\ &= -\Omega_0 \sin(\Omega_0 t), \end{aligned}$$

as was to have been shown.

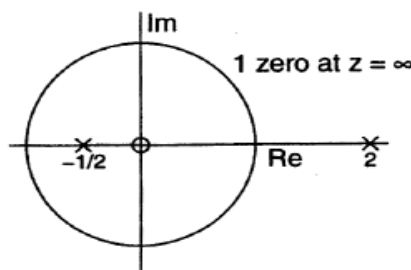
5.8

(a)

$$\begin{aligned}
 y[n] &= \frac{3}{2}y[n-1] + y[n-2] + x[n-1] \\
 Y(z) &= \frac{3}{2}z^{-1}Y(z) + z^{-2}Y(z) + z^{-1}X(z)
 \end{aligned}$$

Therefore,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z^{-1}}{1 - \frac{3}{2}z^{-1} - z^{-2}} = \frac{z^{-1}}{(1 - 2z^{-1})(1 + \frac{1}{2}z^{-1})}, \quad |z| > 2$$



(b)

$$\begin{aligned}
 H(z) &= \frac{z^{-1}}{(1 - 2z^{-1})(1 + \frac{1}{2}z^{-1})} = \frac{A}{(1 - 2z^{-1})} + \frac{B}{(1 + \frac{1}{2}z^{-1})}, \quad |z| > 2 \\
 A &= \left. \frac{z^{-1}}{(1 + \frac{1}{2}z^{-1})} \right|_{z^{-1} = \frac{1}{2}} = \frac{2}{5} \\
 B &= \left. \frac{z^{-1}}{(1 - 2z^{-1})} \right|_{z^{-1} = -2} = -\frac{2}{5}
 \end{aligned}$$

$$h[n] = \frac{2}{5} \left[(2)^n - \left(-\frac{1}{2}\right)^n \right] u[n]$$

(c) Use ROC of $\frac{1}{2} < |z| < 2$ since the ROC must include $|z| = 1$ for a stable system.

$$h[n] = -\frac{2}{5}(2)^n u[-n-1] - \frac{2}{5} \left(-\frac{1}{2}\right)^n u[n]$$

5.12

(a) Yes. The poles $z = \pm j(0.9)$ are inside the unit circle so the system is stable.(b) First, factor $H(z)$ into two parts. The first should be minimum phase and therefore have all its poles and zeros inside the unit circle. The second part should contain the remaining poles and zeros.

$$H(z) = \underbrace{\frac{1 + 0.2z^{-1}}{1 + 0.81z^{-2}}}_{\text{minimum phase}} \underbrace{\frac{1 - 9z^{-2}}{1}}_{\substack{\text{poles \& zeros} \\ \text{outside unit circle}}}$$

Allpass systems have poles and zeros that occur in conjugate reciprocal pairs. If we include the factor $(1 - \frac{1}{9}z^{-2})$ in both parts of the equation above the first part will remain minimum phase and the second will become allpass.

$$\begin{aligned}
 H(z) &= \frac{(1 + 0.2z^{-1})(1 - \frac{1}{9}z^{-2})}{1 + 0.81z^{-2}} \cdot \frac{1 - 9z^{-2}}{1 - \frac{1}{9}z^{-2}} \\
 &= H_1(z)H_{ap}(z)
 \end{aligned}$$

5.15

(a) $h[n]$ is symmetric about $n = 1$.

$$\begin{aligned} H(e^{j\omega}) &= 2 + e^{-j\omega} + 2e^{-2j\omega} \\ &= e^{-j\omega}(2e^{j\omega} + 1 + 2e^{-j\omega}) \\ &= (1 + 4\cos\omega)e^{-j\omega} \\ A(\omega) &= 1 + 4\cos\omega, \alpha = 1, \beta = 0 \end{aligned}$$

Generalized Linear phase but not Linear Phase since $A(\omega)$ is not always positive.

(b) This sequence has no even or odd symmetry, so it does not possess generalized linear phase.

(c) $h[n]$ is symmetric about $n = 1$.

$$\begin{aligned} H(e^{j\omega}) &= 1 + 3e^{-j\omega} + e^{-2j\omega} \\ &= e^{-j\omega}(e^{j\omega} + 3 + e^{-j\omega}) \\ &= (3 + 2\cos\omega)e^{-j\omega} \\ A(\omega) &= 3 + 2\cos\omega, \alpha = 1, \beta = 0 \end{aligned}$$

Generalized Linear phase & Linear Phase.

(d) $h[n]$ has even symmetry.

$$\begin{aligned} H(e^{j\omega}) &= 1 + e^{-j\omega} \\ &= e^{-j(1/2)\omega}(e^{j(1/2)\omega} + e^{-j(1/2)\omega}) \\ &= 2\cos(\omega/2)e^{-j(1/2)\omega} \\ A(\omega) &= 2\cos(\omega/2), \alpha = \frac{1}{2}, \beta = 0 \end{aligned}$$

Generalized Linear Phase but not Linear Phase since $A(\omega)$ is not always positive.

(e) $h[n]$ has odd symmetry.

$$\begin{aligned} H(e^{j\omega}) &= 1 - e^{-2j\omega} \\ &= e^{-j\omega}(e^{j\omega} - e^{-j\omega}) \\ &= e^{-j\omega}2j\sin\omega \\ &= (2\sin\omega)e^{-j\omega+j\frac{\pi}{2}} \\ A(\omega) &= 2\sin\omega, \alpha = 1, \beta = \frac{\pi}{2} \end{aligned}$$

Generalized Linear Phase but not Linear Phase since $A(\omega)$ is not always positive.

Matlab

Problem:

Plot the frequency response and group delay of

$$H(z) = \frac{0.001836(1+z^{-1})^4}{(1-1.5548z^{-1}+0.6493z^{-2})(1-1.4996z^{-1}+0.8482z^{-2})}$$

Answer:

```
K = input('Number of frequency points=')
w = 0:pi/(K-1):pi;

n1 = [1 1];
num = 0.001836*conv(conv(conv(n1,n1),n1),n1);
d1 = [1 -1.5548 0.6493];
d2 = [1 -1.4996 0.8482];
den = conv(d1,d2);

h = freqz(num,den,w);
grd = grpdelay(num,den,w);

subplot(3,1,1);
plot(w,angle(h));
xlabel('Frequency \omega')
ylabel('Radian')
title('Phase spectrum')

subplot(3,1,2);
plot(w,abs(h));
xlabel('Frequency \omega')
ylabel('Amplitude')
title('Magnitude spectrum')

subplot(3,1,3);
plot(w,grd);
xlabel('Frequency \omega')
ylabel('Samples')
title('Group Delay')
```

K=256;

